MAX SCHNEIDER, PhD.

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Predictive Models | Seismic Hazards | Machine Learning | Data Visualization | Research Expertise

Applied statistician brings 15+ years of expertise leading teams to build right-sized statistical and machine learning models for seismic hazards. Expert in designing metrics, evaluating model performance, and visualizing risk under uncertainty using real and synthetic data. Frames and solves ambiguous technical problems while quickly acquiring concepts in diverse fields. Proven leadership of multidisciplinary teams to drive data-informed decision-making across engineering, research, and client-focused domains.

- ***Programming & Tools***: Python, R, SQL, C++, bash, Git, HPC/Supercomputing, GIS, Shiny, ggplot2
- ***Statistics & ML***: Bayesian modeling, (generalized) linear regression, time series, spatial/spatiotemporal methods, clustering, boosted regression, uncertainty quantification, model validation, A/B testing
- ***Technical Communication***: client engagement, data visualization, map design, scientific writing
- ***Team Leadership***: project vision, management and execution across disciplines, skillsets, and countries (proficient in Spanish, German, Russian)

EDUCATION

University of Washington, Seattle *September 2016 – December 2021*
Doctor of Philosophy in Statistics

University of Potsdam (Potsdam, Germany) *October 2012 – February 2016*
Master of Science in Mathematics: Mathematical Modeling and Data Analysis

University of California, Los Angeles *September 2007 – June 2011*
Bachelor of Science in Statistics and Human Complex Systems

WORK EXPERIENCE

Independent Statistical Consultant May 2026 –

- Italian Inst. of Geophysics and Volcanology. Develop operational forecasting and reporting systems based on emergency management needs for national Italian earthquake forecasts
- Calif. Geol. Survey. Design human-centered color palettes and legends to improve California's state seismic hazard map

U.S. Geological Survey May 2022 – May 2026
Research Statistician (Mendenhall Postdoctoral Fellow) *Moffett Field, CA*

- Developed novel forecast evaluation metrics and frameworks to assess USGS aftershock forecast models, influencing future model deployments
- Parallelized code for high-performance computing to efficiently calculate and evaluate millions of probabilistic forecasts
- Engaged with stakeholders from engineering, critical infrastructure (utility, transportation), emergency managers, public health and public information to improve earthquake forecasting in the U.S., Mexico and El Salvador
- Designed and tested actionable maps, GIS systems and interactive software for public-facing aftershock forecasts
- Mentored interns and lead 20+ member international research team

German Research Center for Geosciences August 2020 – May 2022
Statistical Consultant with Dr. Fabrice Cotton *Potsdam, Germany*

- Optimized colors and data classifications for natural hazard maps to improve hazard perception for non-technical users
- Designed seismic hazard maps for Germany and evaluated how well they communicated key targets in user study

University of Washington Department of Statistics September 2016 – December 2021
Graduate Research Assistant with Dr. Peter Guttorp *Seattle, WA*

- Built first-of-its-kind model of earthquake occurrence in the Pacific Northwest with spatiotemporal point processes, creating Bayesian method to estimate parameters and quantify uncertainty even with limited data
- Developed automated classification of different types of earthquake clusters to facilitate seismic source modelling for more complex tectonic environments than allowed by prior methods
- Designed and A/B-tested several approaches to visualize uncertainty for forecast maps, showing how one method can improve communication of aftershock forecasts to non-technical audiences

UW Center for Statistics and the Social Sciences September 2019 – June 2020
Statistical Consultant *Seattle, WA*

- Provided statistical advising to dozens of clients across the social and health sciences, improving their research with experiment/survey design, machine learning, linear, multilevel, spatial and time series regression

Humboldt University of Berlin, Geography Institute April 2015 – September 2016
Research Assistant with Drs. Tobias Kümmerle and Tobia Lakes *Berlin, Germany*

- Built machine learning models (boosted regression) to efficiently identify predictors of land use changes for the first time for finely-gridded spatial data across European continent *(Levers et al., 2018)*
- Mapped spatially-varying models in GIS to reveal relationships between predictor and response variables that were not previously studied

UCLA Department of Atmospheric and Oceanic Sciences March 2008 – January 2011
Research Assistant with Dr. Jochen Stutz *Los Angeles, CA*

- Conducted fieldwork and computational modelling to determine whether a spectrometry instrument could sufficiently measure the concentration of ozone and other compounds in the stratosphere *(Stutz et al., 2011)*

SELECTED AWARDS AND MEMBERSHIPS

- USGS STAR Awards for innovation and leadership (three awards in February 2023, March 2023, February 2024)
- USGS-Bureau of Humanitarian Assistance grant for post-doctoral research project ($411,606, 2022-2025)
- Seismological Society of America Annual Meeting Student Presentation Award (April 2021)
- USGS Earthquake Hazards Program External Grant ($99,863, 2020-2021)
- Member of the Earthquake Eng. Research Institute, Seism. Society of America, American Statistical Association

SELECTED PUBLICATIONS

Statistical and Computational Methods for Earth Sciences

1. **Schneider, M.**, Barall, M., Guttorp, P., Hardebeck, J., Michael, A., Page, M., and van der Elst, N. Bayesian ETAS modelling for the Pacific Northwest: Uncovering effects of tectonic regimes, regional differences and swarms on aftershock parameters, *Bulletin of the Seismological Society of America*, 5 , 2219-2236, 2025.
2. **Schneider, M.** and Guttorp, P. What do we know without the catalog? Eliciting prior beliefs from experts for aftershock models, *The Seismic Record*, 4, 259-267, 2024.
3. **Schneider, M.**, Flury, H., Guttorp, P. and Wright, A. Earthquake catalog processing and swarm identification for the Pacific Northwest, *Seismological Research Letters*, 94, 2500–2513, 2023.
4. Levers, C., **Schneider, M.**, Prishchepov, A., Estel, S. and Kümmerle, T. Spatial variation in determinants of agricultural land abandonment in Europe. *Science of the Total Environment*, 644, 95-111, 2018.
5. Stutz, J., Thomas, J.L., Hurlock, S.C., **Schneider, M.**, Von Glasow, R., ... Longpath DOAS obser-vations of surface BrO at Summit, Greenland. *Atmospheric Chemistry and Physics,* 11, 9899-9910, 2011.

Human-Centered Visualization of Risk

6. **Schneider, M.**, Wein, A., McBride, S.K., van der Elst, N., Becker, J., Castro, R., Diaz, M., ... Meet the people where they are: Assessing user needs for aftershock forecast products in El Salvador, Mexico and the United States, *International Journal of Disaster Risk Reduction*, 125, 2025.
7. **Schneider, M.**, Cotton, F., and Schweizer, P.-J. Criteria-based visualization design for hazard maps, *Natural Hazards and Earth System Sciences,* 23, 2505-2521, 2023.
8. **Schneider, M.**, McDowell, M., Guttorp, P., Steel, E.A., and Fleischhut, N. Effective uncertainty visualization for aftershock forecast maps, *Natural Hazards and Earth System Sciences*, 22, 1499-1518, 2022.